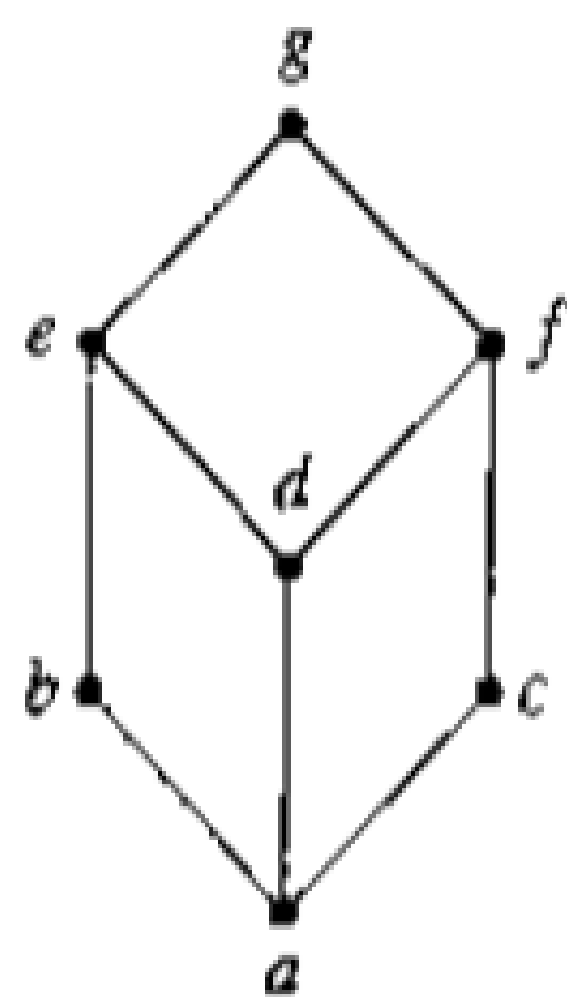


Discrete Mathematics — Final Examination (Paper B)

|                |                                                                                                                                                                                |    |    |    |   |    |    |    |    |    |  |  |    |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---|----|----|----|----|----|--|--|----|
| 考试<br>注意<br>事项 | 一、学生参加考试须带学生证或学院证明，未带者不准进入考场。学生必须按照监考教师指定座位就坐。<br>二、书本、参考资料、书包等与考试无关的东西一律放到考场指定位置。<br>三、学生不得另行携带、使用稿纸，要遵守《北京邮电大学考场规则》，有考场违纪或作弊行为者，按相应规定严肃处理。<br>四、学生必须将答题内容做在试题答卷上，做在草稿纸上一律无效。 |    |    |    |   |    |    |    |    |    |  |  |    |
| 考试课程           | 离散数学                                                                                                                                                                           |    |    |    |   |    |    |    |    |    |  |  |    |
| 题号             | 一                                                                                                                                                                              | 二  | 三  | 四  | 五 | 六  | 七  | 八  | 九  | 十  |  |  | 总分 |
| 满分             | 10                                                                                                                                                                             | 10 | 10 | 14 | 6 | 10 | 10 | 10 | 10 | 10 |  |  |    |
| 得分             |                                                                                                                                                                                |    |    |    |   |    |    |    |    |    |  |  |    |
| 阅卷教师           |                                                                                                                                                                                |    |    |    |   |    |    |    |    |    |  |  |    |

1. [10 points] Suppose  $A=\{a,b,c,d,e\}$ . Let  $R$  and  $S$  be the relations on  $A$  described by  $R=\{(a,c), (b,c),(c,e), (d,a), (d,b),(e,c)\}$  and  $S=\{(a,b),(a,c),(c,d),(d,e),(e,a)\}$ . Use Warshall's algorithm to compute the transitive closure of  $S\circ R$ .
2. [10 points] Answer these questions for the partial order represented by this Hasse diagram.



- a) Compute  $LUB(\{b,c\})$ .

b) Compute  $GLB(\{f, b\})$ .

c) Is the poset a lattice? Explain your answer.
3. [10 points] Prove that if  $(G,*)$  and  $(G',*')$  are Abelian groups, then  $(G \times G',*")$  is an Abelian Group.

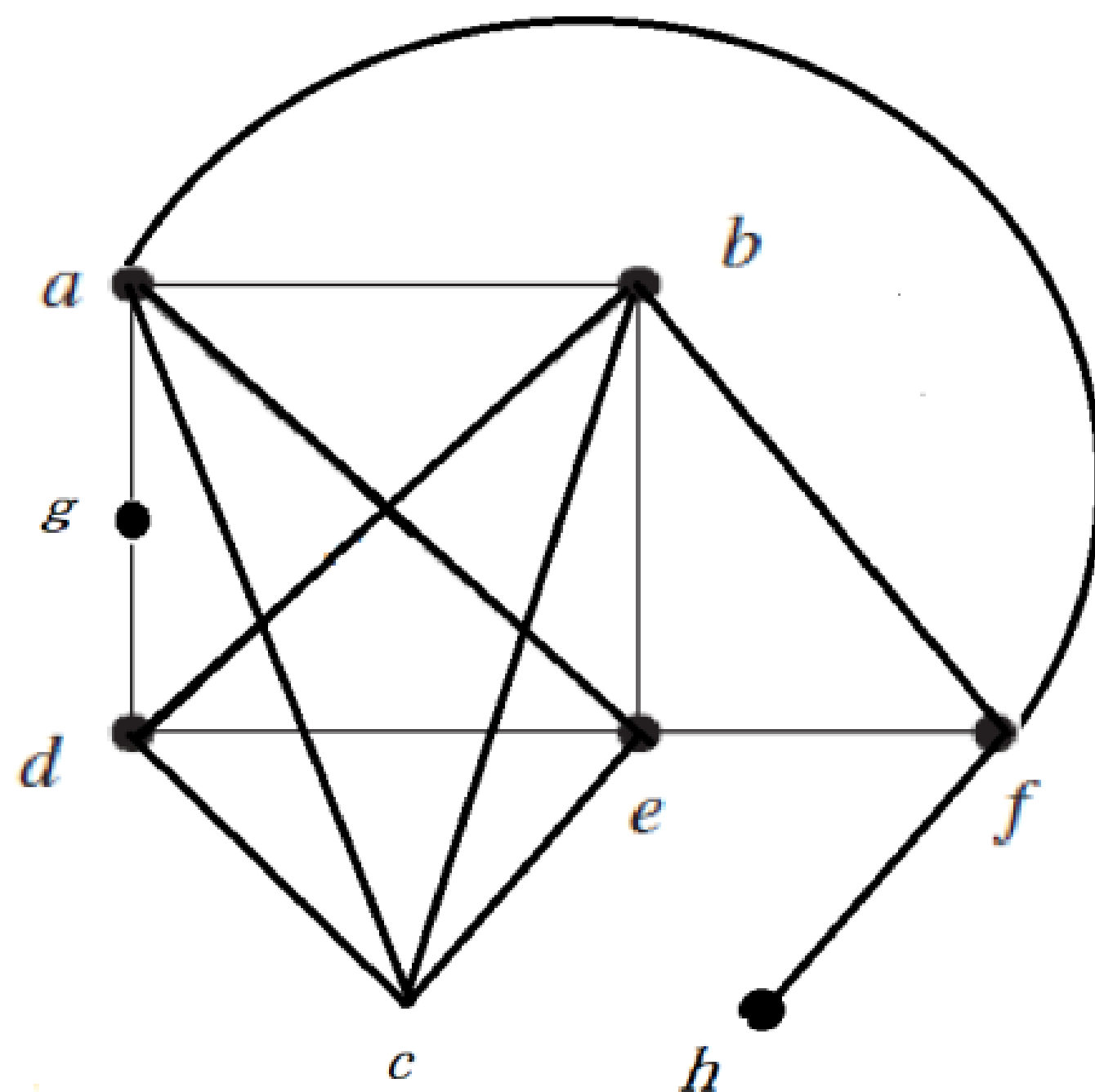


4. [14 points] Let  $m=3$ ,  $n=6$ ,  $H = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$  be a parity check matrix

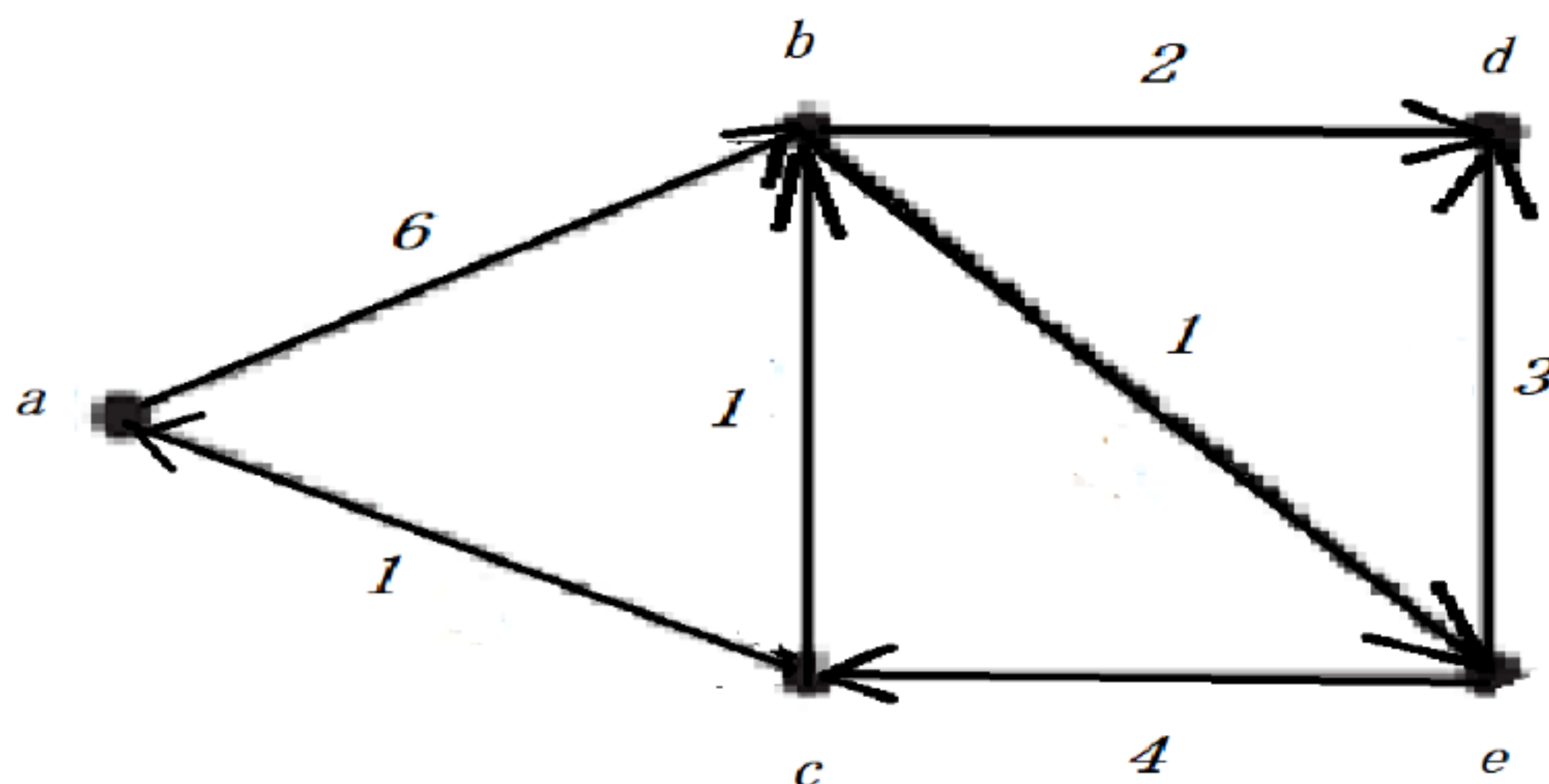
- Determine the  $(3, 6)$  group code  $e_H: B^3 \rightarrow B^6$ .
- Find the minimal distance of  $e_H$ .
- How many errors will  $e_H$  detect?
- Decode the following words relative to a maximum likelihood decoding function associated with  $e_H$ .

(1) 101011      (2) 111011      (3) 000111

5. [6 points] Determine whether the given graphs have an Euler circuit or a Hamilton circuit. If not, determine whether the given graphs have an Euler path or a Hamilton path. And determine whether the given graphs are planar.

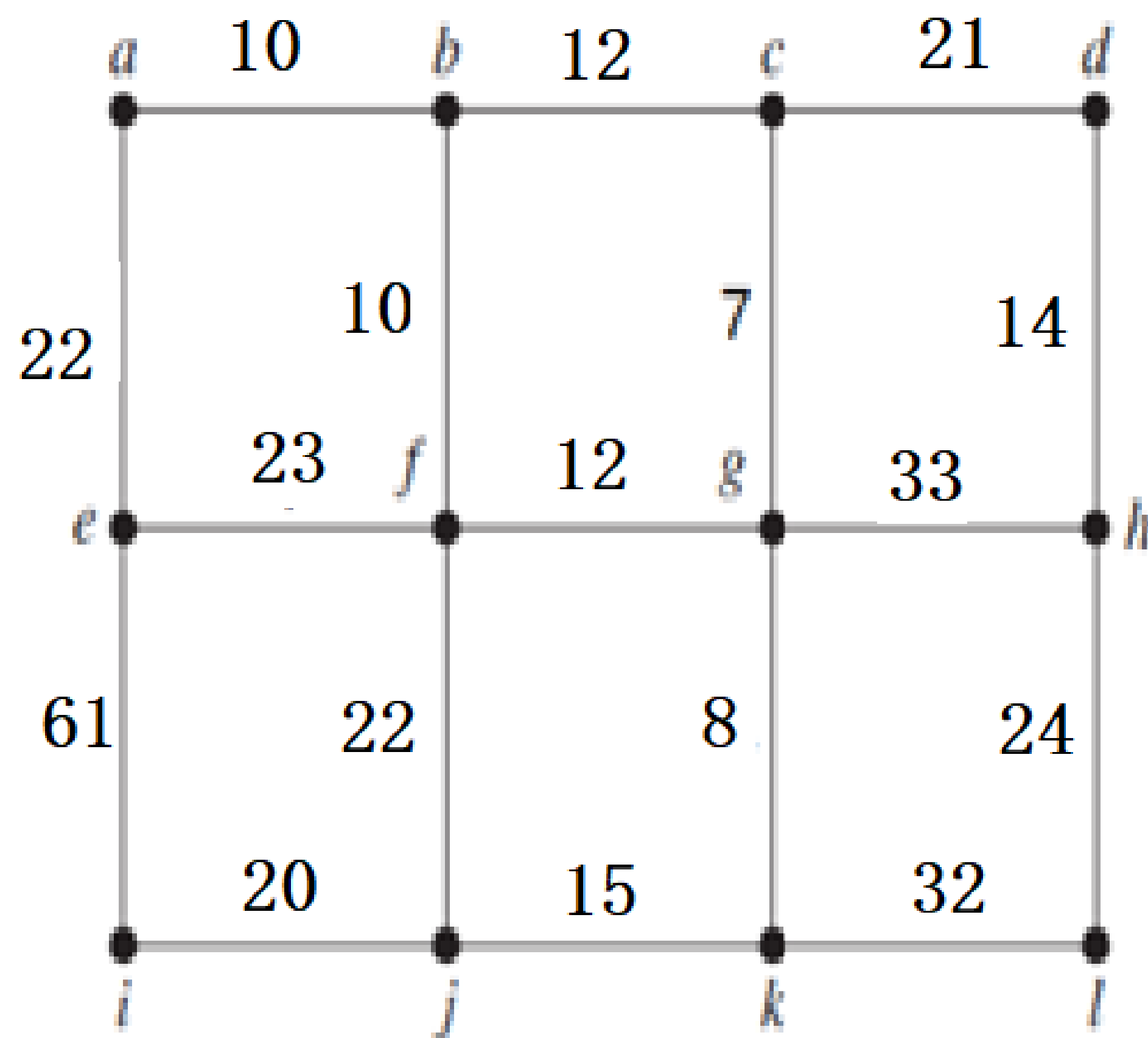


6. [10 points] Finding the shortest path length between any two vertices using distance matrix.

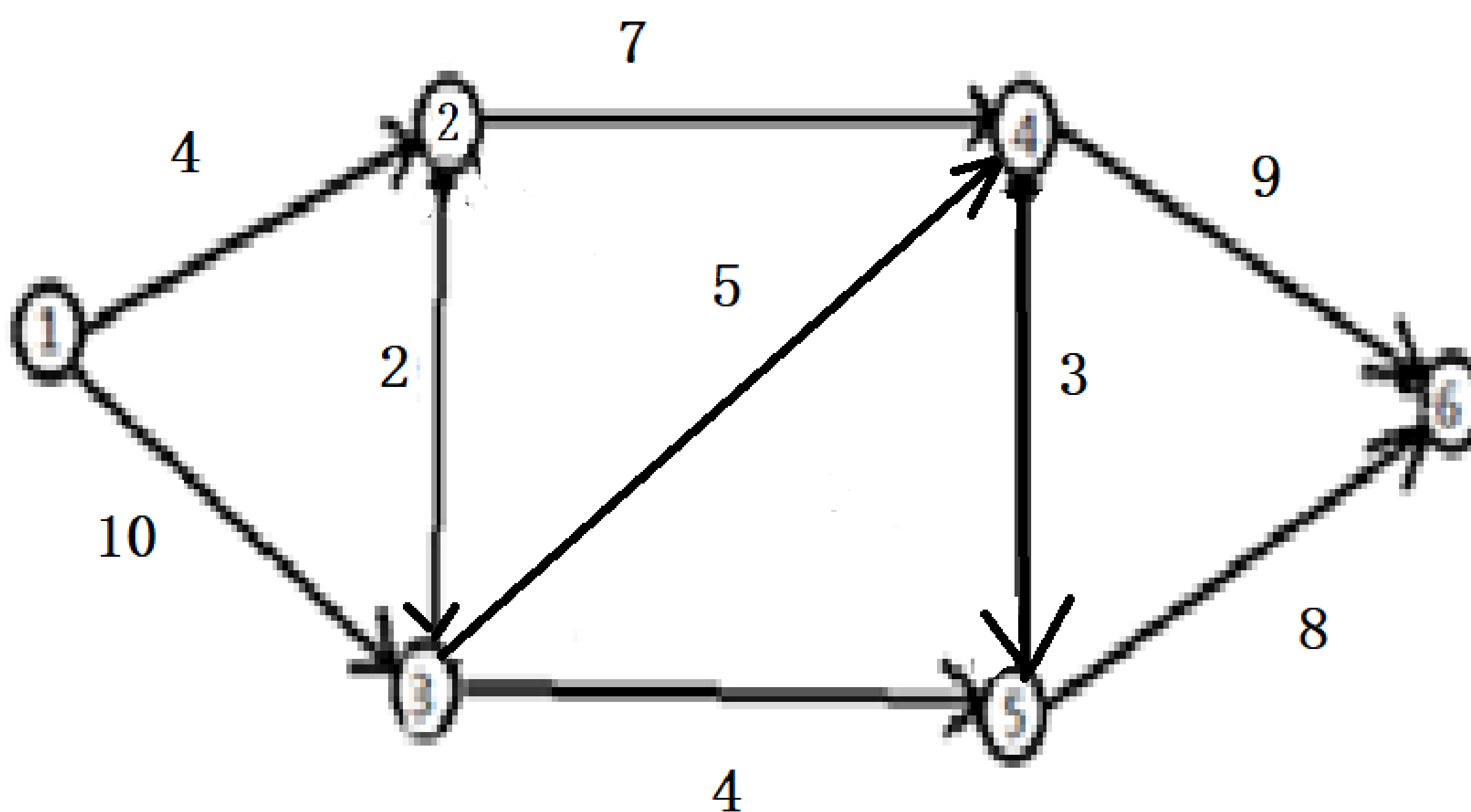




7. [10 points] Use Kruskal's algorithm to design a minimum-cost communications network connecting all the computers represented by the graph in next Figure.



8. [10 points] Find a maximum flow in the given network by using the labeling algorithm. And find a minimum cut of this network. (Please give out the labeling graph of every flow.)



9. [10 points] Find the solution to the recurrence relations  

$$a_n = 5a_{n-1} - 4a_{n-2}, a_0 = 1, a_1 = 9.$$
10. [10 points] Set up a generating function and use it to find the number of ways in which eleven identical coins can be put in three distinct envelopes (labeled A, B, C) if envelope A has at least three coins in it.



1. [10 points] Suppose  $A = \{a, b, c, d, e\}$ . Let  $R$  and  $S$  be the relations on  $A$  described by  $R = \{(a, c), (b, c), (c, e), (d, a), (d, b), (e, c)\}$  and  $S = \{(a, b), (a, c), (c, d), (d, e), (e, a)\}$ . Use Warshall's algorithm to compute the transitive closure of  $S \circ R$ .

$$S \circ R = \{(a, d), (b, d), (c, a), (d, b), (d, c), (e, d)\}$$

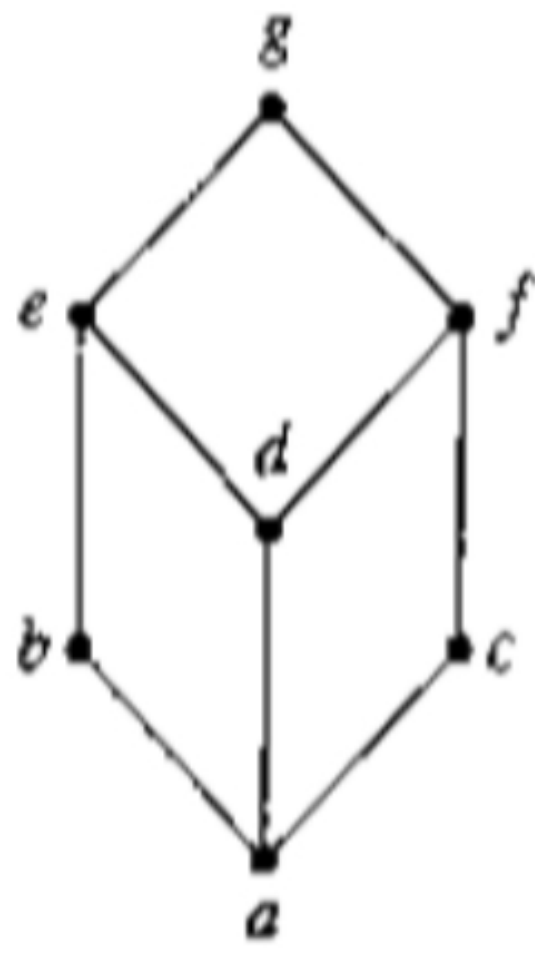
$$W_0 = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \quad W_1 = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \quad W_3 = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$W_4 = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix} = W_5$$



2. [10 points] Answer these questions for the partial order represented by this Hasse diagram.



- a) Compute  $\text{LUB}(\{b, c\})$ .
- b) Compute  $\text{GLB}(\{f, b\})$ .
- c) Is the poset a lattice? Explain your answer.

a)  $\text{LUB}(\{b, c\}) = \{g\}$

b)  $\text{GLB}(\{f, b\}) = \{a\}$

c) 是, 任意子集都能找到上界:  $\{g\}$ , 下界:  $\{a\}$ ,  
所以  $\text{GLB}$  和  $\text{LUB}$  一定存在.



3. [10 points] Prove that if  $(G, *)$  and  $(G', *')$  are Abelian groups, then  $(G \times G', *'')$  is an Abelian Group.

$\therefore (G, *)$  and  $(G', *)$  are Abelian,

$\therefore a, b \in G, c, d \in G', a * b = b * a, c *' d = d *' c$

$\therefore (G \times G', *'') :$

$$\begin{aligned} & (a, c) *'' (b, d) \\ &= (a * b, c *' d) \\ &= (b * a, d *' c) \\ &= (b, d) *'' (a, c) \end{aligned}$$

$\therefore (G \times G', *'') \text{ is Abelian.}$



4. [14 points] Let  $m=3$ ,  $n=6$ ,  $H = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$  be a parity check matrix

- Determine the  $(3, 6)$  group code  $e_H: B^3 \rightarrow B^6$ .
- Find the minimal distance of  $e_H$ .
- How many errors will  $e_H$  detect?
- Decode the following words relative to a maximum likelihood decoding function associated with  $e_H$ .  
(1) 101011      (2) 111011      (3) 000111

a)

$$H' = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{array}{ll} e(000) = 000000 & e(100) = 100110 \\ e(001) = 001011 & e(101) = 101101 \\ e(010) = 010101 & e(110) = 110011 \\ e(011) = 011110 & e(111) = 111000 \end{array}$$

b) minimal distance : 3

c)  $e$  can detect 2 errors

d)  $\min \{ \delta(101011), e(b) \} = 1$

$$d(101011) = 011$$

$$\min \{ \delta(111011), e(b) \} = 1$$

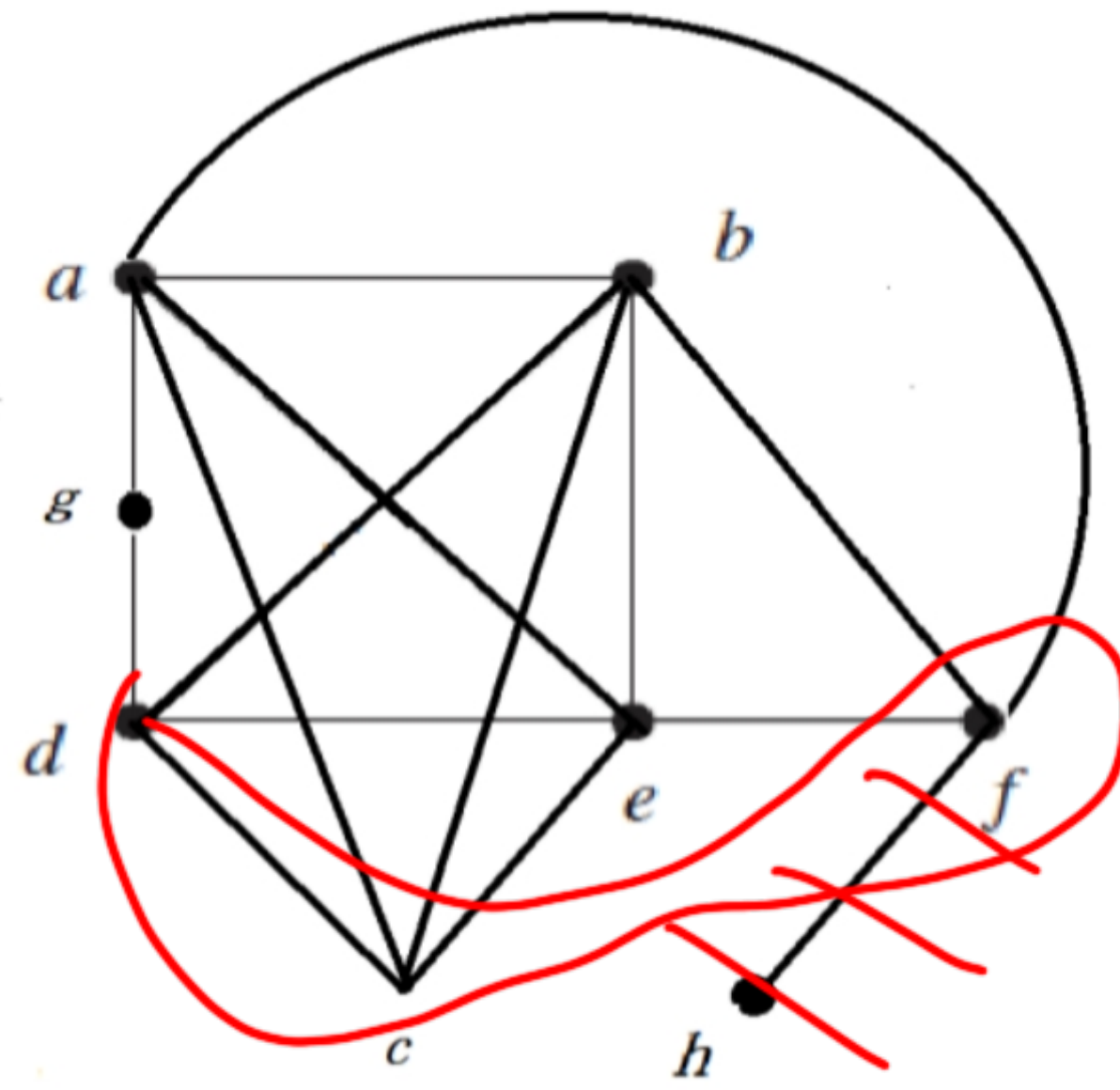
$$d(111011) = 110$$

$$\min \{ \delta(000111), e(b) \} = 2$$

$$d(000111) = 001 / 010 / 100$$



5. [6 points] Determine whether the given graphs have an Euler circuit or a Hamilton circuit. If not, determine whether the given graphs have an Euler path or a Hamilton path. And determine whether the given graphs are planar.



$\because$  h 点度数为 1,

$\therefore$  不含欧拉回路和汉米尔顿回路

$\because$  a, b, e 度均为 5,  $\therefore$  不含欧拉通路

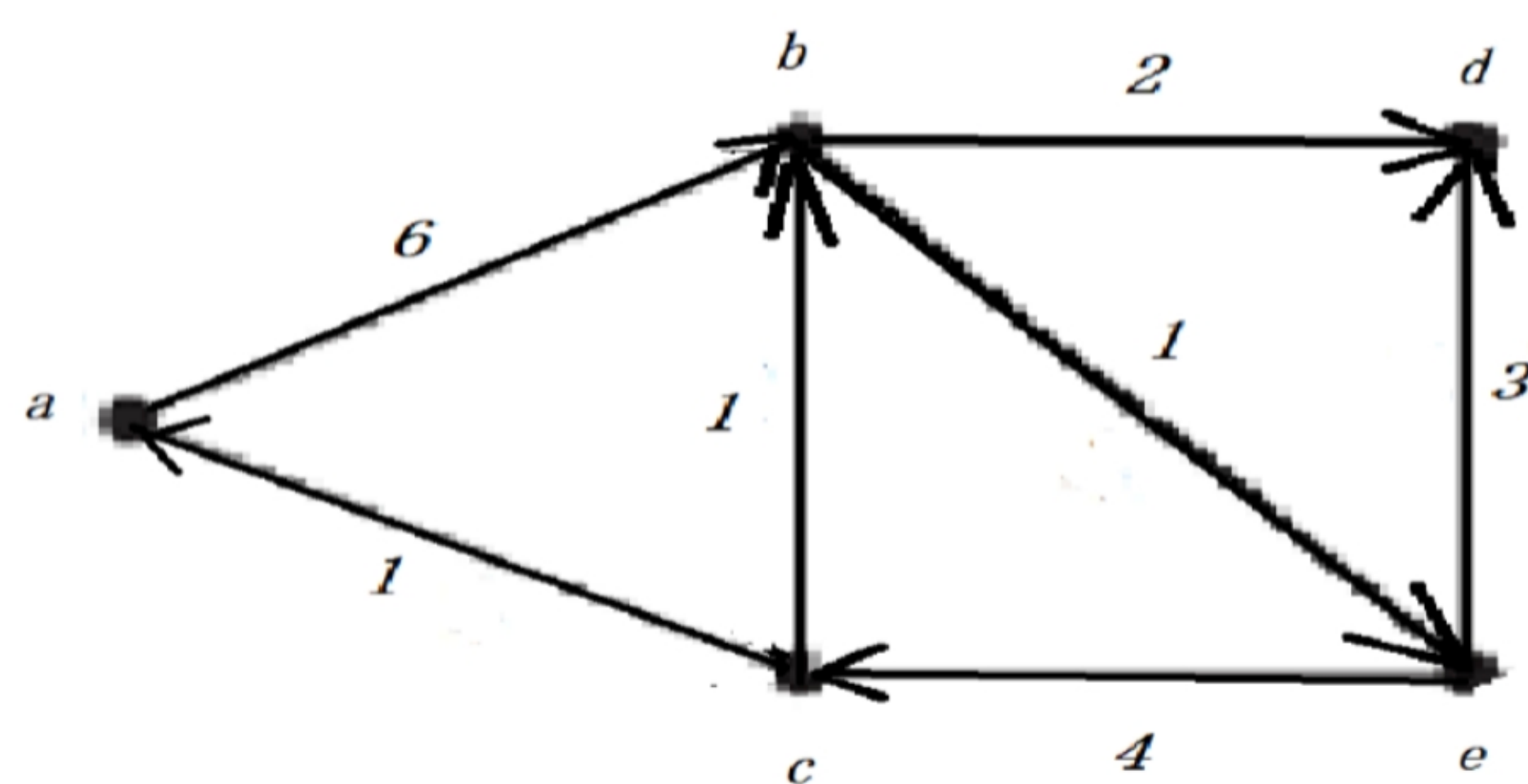
汉米尔顿通路: h f a g d c e b

$\{d, c, f\}$  和  $\{a, b, e\}$  可以构成  $K_{3,3}$

$\therefore$  不是平面图



6. [10 points] Finding the shortest path length between any two vertices using distance matrix.

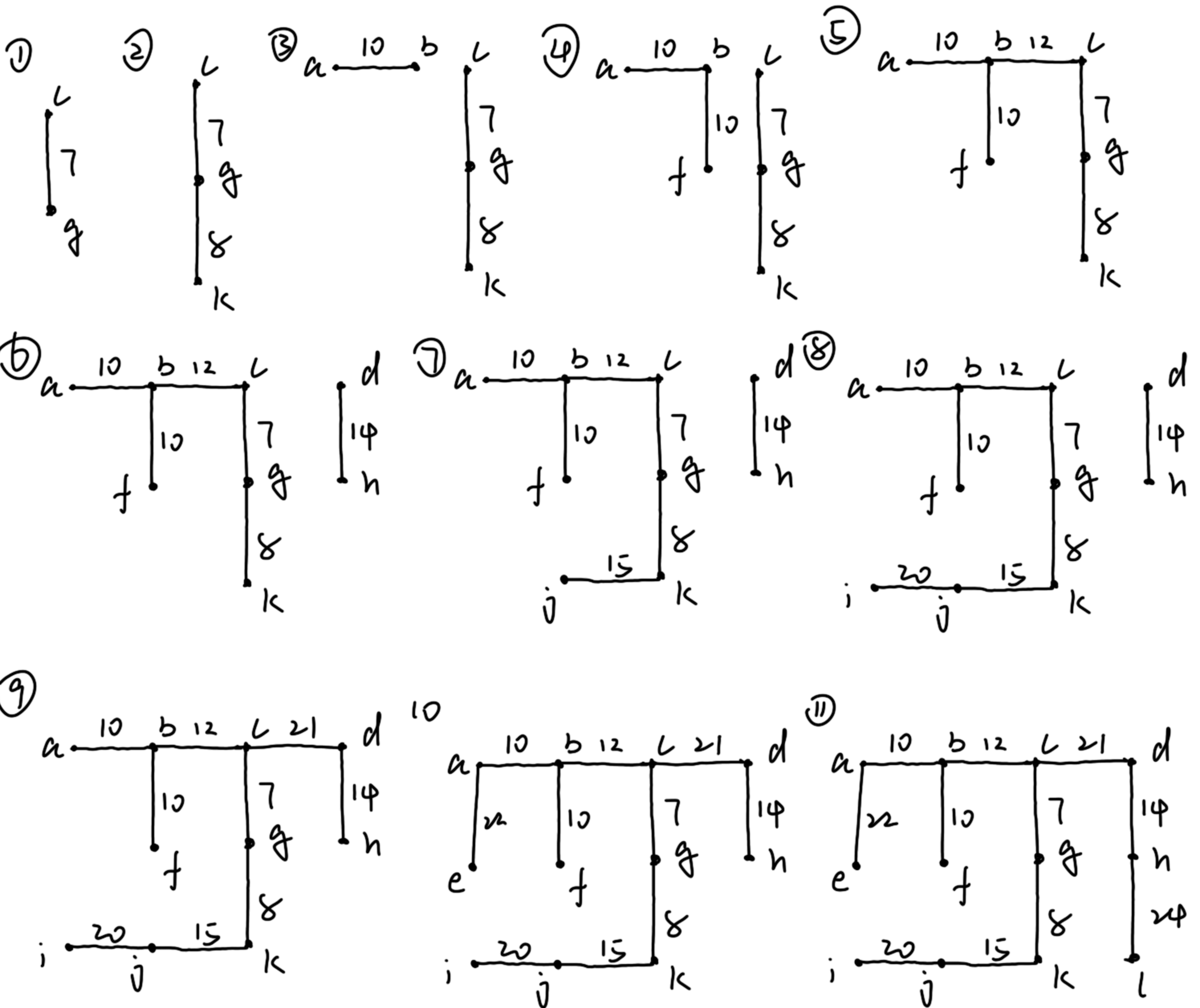
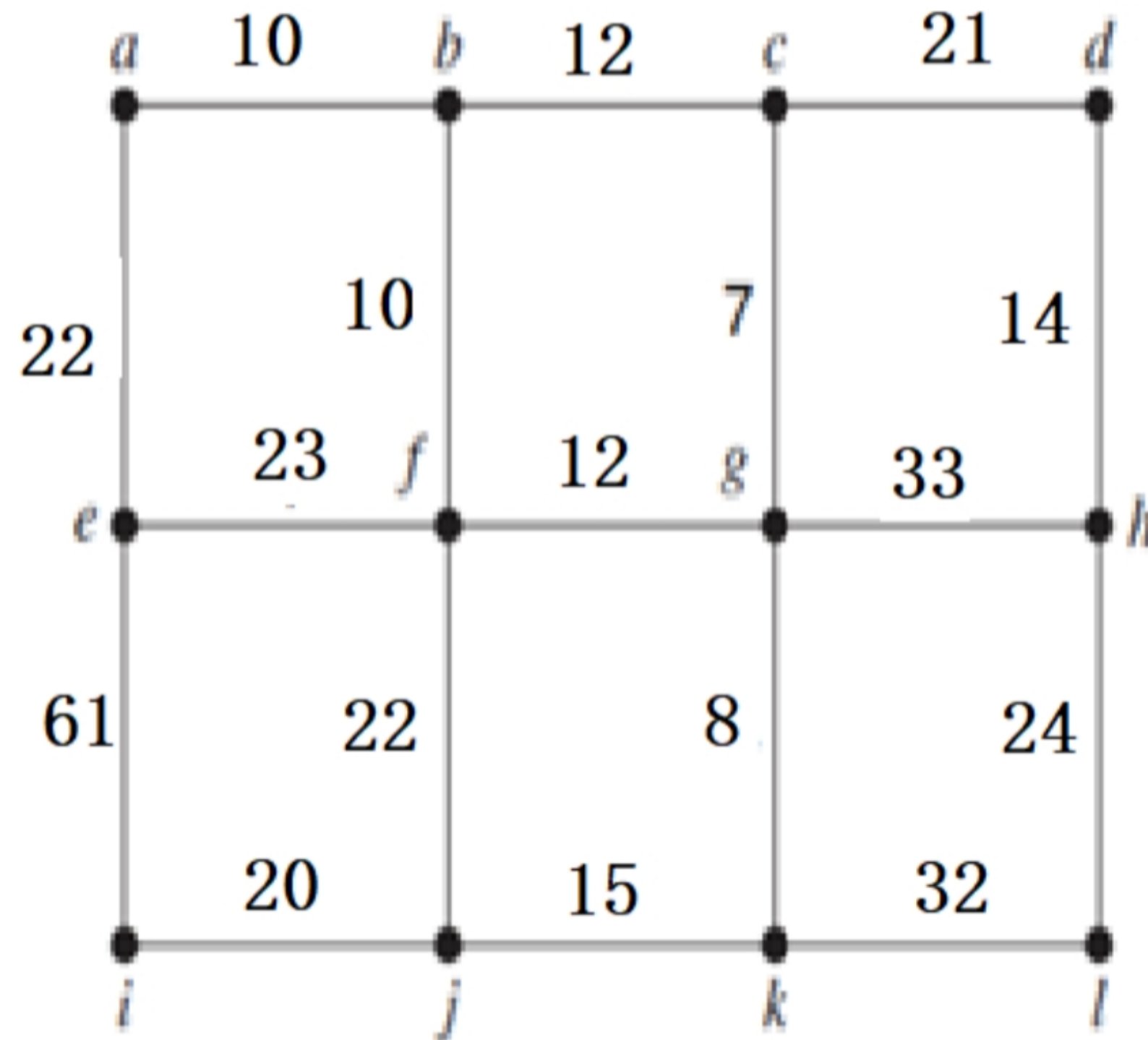


{a, b, c, e} 可以构成强连通分支.

$$\begin{bmatrix}
 12 & 6 & 11 & 8 & 7 \\
 6 & 6 & 5 & 2 & 1 \\
 1 & 1 & 6 & 3 & 2 \\
 0 & 0 & 0 & 0 & 0 \\
 5 & 5 & 4 & 3 & 6
 \end{bmatrix}$$

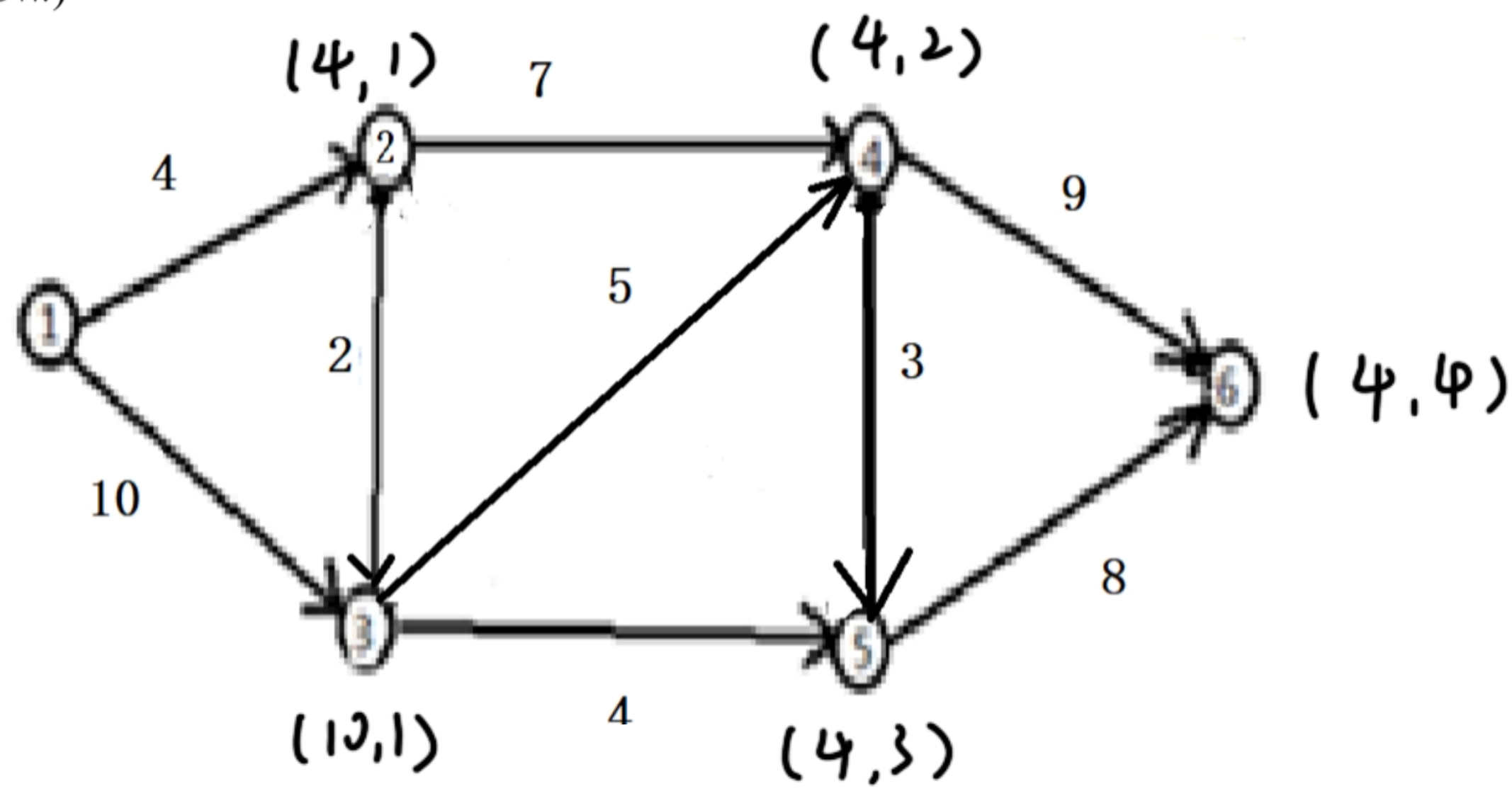


7. [10 points] Use Kruskal's algorithm to design a minimum-cost communications network connecting all the computers represented by the graph in next Figure.



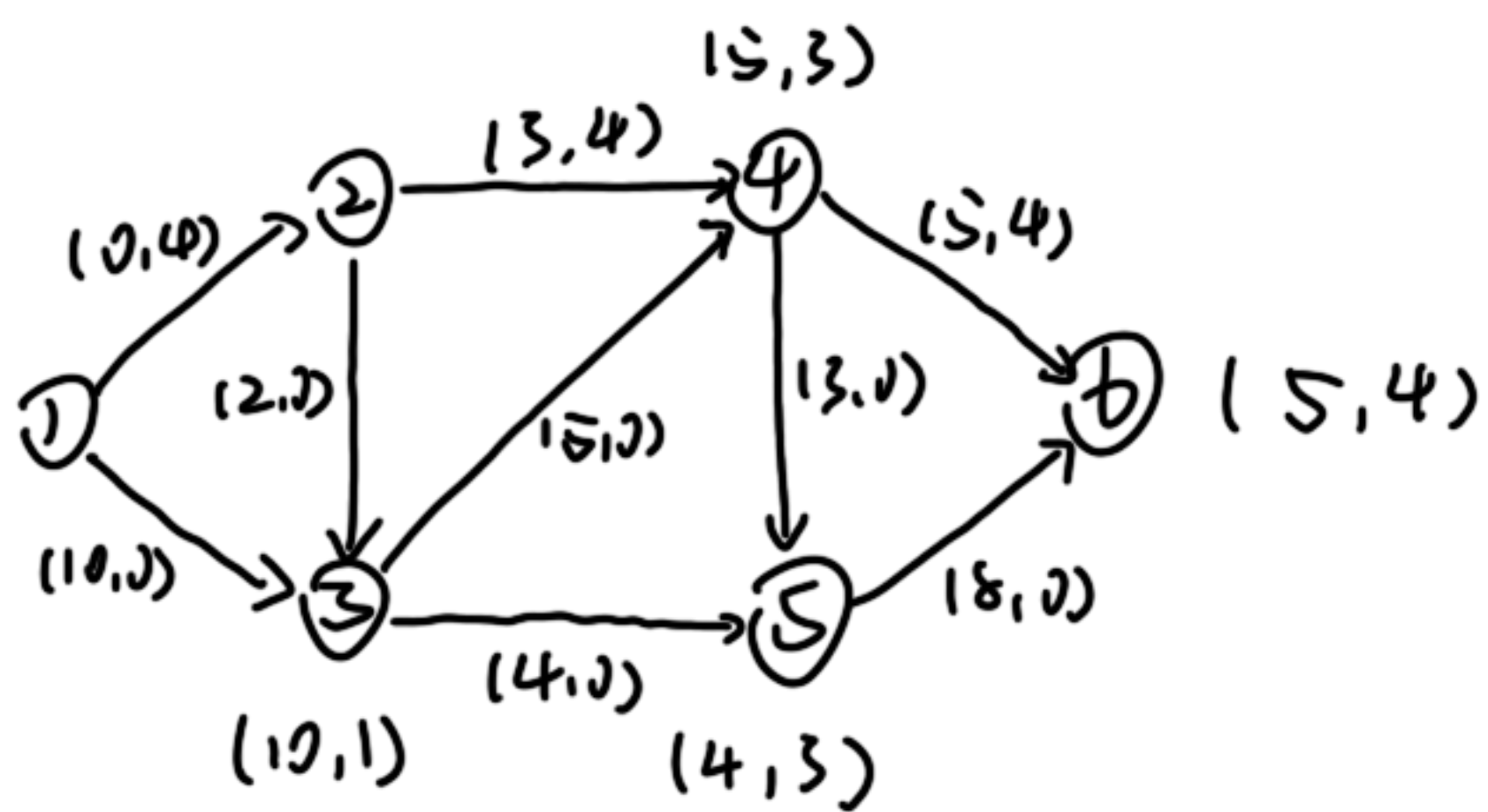


8. [10 points] Find a maximum flow in the given network by using the labeling algorithm. And find a minimum cut of this network. (Please give out the labeling graph of every flow.)

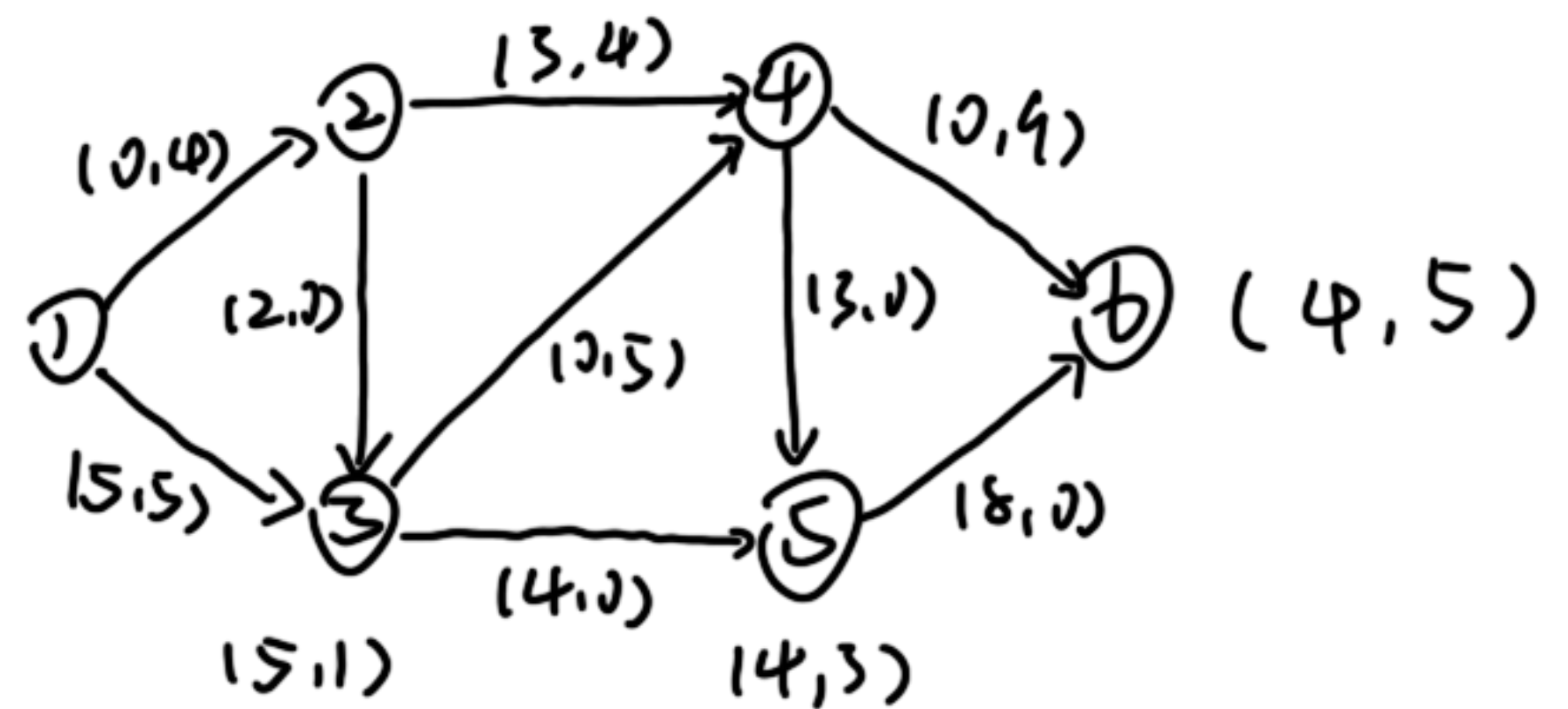


$$N_1 = \{2, 3\} \quad N_2 = \{4, 5\} \quad N_3 = \{6\}$$

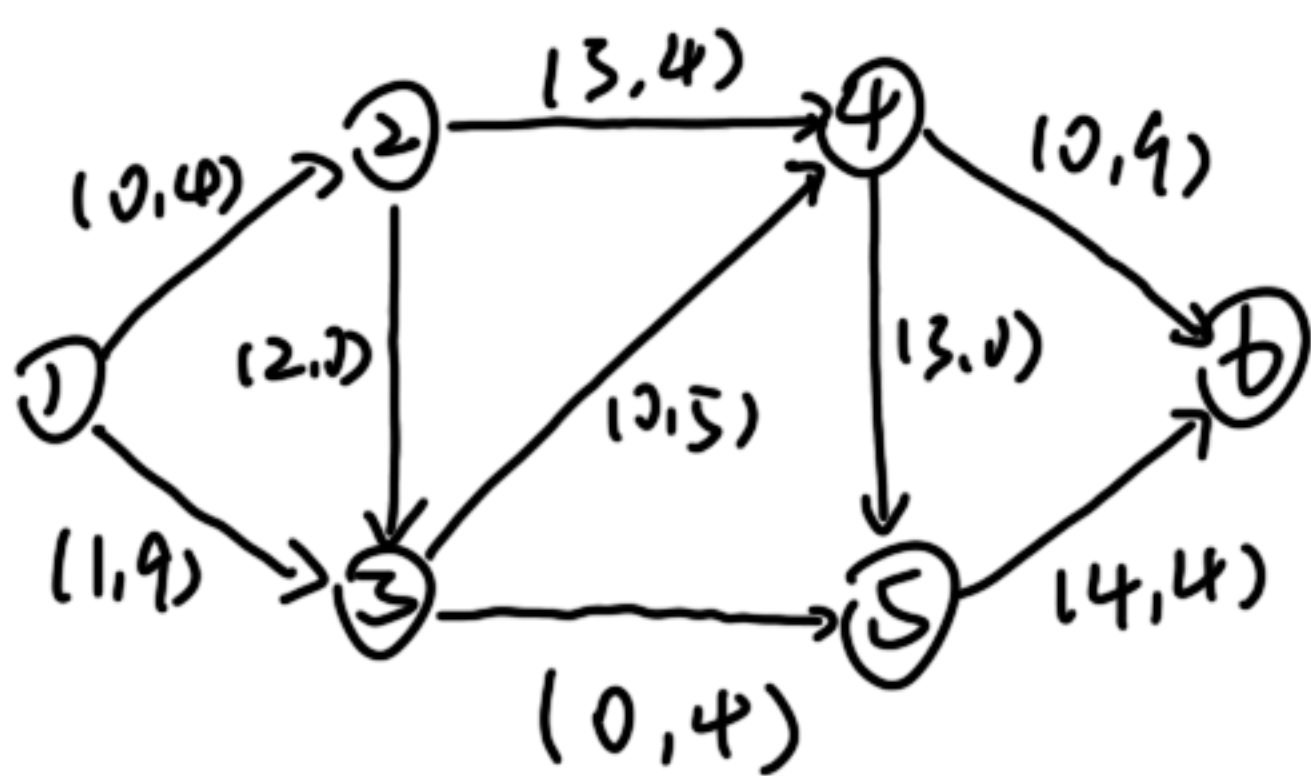
$$1-2-4-6 \quad \text{flow} = 4$$



$$1-3-4-6 \quad \text{flow} = 5$$



$$1-3-5-6 \quad \text{flow} = 4$$



$$\text{无路径} \quad \text{max flow} = 13$$



9. [10 points] Find the solution to the recurrence relations

$$a_n = 5a_{n-1} - 4a_{n-2}, a_0 = 1, a_1 = 9.$$

$$h(x) = \sum_{k=0}^{\infty} a_k x^k \quad 5x h(x) = \sum_{k=1}^{\infty} 5a_{k-1} x^k \quad 4x^2 h(x) = \sum_{k=2}^{\infty} 4a_{k-2} x^k$$

$$h(x) - 5x h(x) + 4x^2 h(x)$$

$$= \sum_{k=2}^{\infty} (a_k - 5a_{k-1} + 4a_{k-2}) x^k + a_0 + a_1 x - 5a_0 x$$

$$= 1 + 4x$$

$$(1 - 5x + 4x^2) h(x) = 1 + 4x$$

$$h(x) = \frac{1+4x}{(1-4x)(1-x)}$$

$$= \frac{8}{3} \cdot \frac{1}{1-4x} - \frac{5}{3} \cdot \frac{1}{1-x}$$

$$= \frac{8}{3} \cdot \sum_{n=0}^{\infty} (2x)^n - \frac{5}{3} \sum_{n=0}^{\infty} x^n$$

$$= \sum_{n=0}^{\infty} \left( \frac{8}{3} \cdot 2^n - \frac{5}{3} \right) x^n$$

$$\therefore a_n = \frac{8}{3} \cdot 2^n - \frac{5}{3}, a_0 = 1, a_1 = 9$$



10. [10 points] Set up a generating function and use it to find the number of ways in which eleven identical coins can be put in three distinct envelopes (labeled A, B, C) if envelope A has at least three coins in it.

$$x_1 + x_2 + x_3 = 11 \quad x_1 \geq 3$$

$$(x^3 + x^4 + \dots)(1 + x + \dots)^2$$

$$= x^3(1 + x + \dots)^3$$

$$= x^3 \cdot \frac{1}{(1-x)^3}$$

$$a_{11} = C_6^2 = 15$$